

$$1) a: \begin{aligned} x &= -1 - 8s \\ y &= 16 + 4s \\ z &= -2s \end{aligned}$$

$$p: \begin{aligned} x &= 3 + 4t \\ y &= 14 - 2t \\ z &= 1 + t \end{aligned}$$

$$P = [-1; 16; 0]$$

$$x = 3 - 4 = -1$$

$$y = 14 + 2 = 16$$

$$z = 1 - 1 = 0$$

$$\vec{s}_a = (-8; 4; -2) \quad \vec{s}_p = (4; -2; 1) \quad A = [-1; 16; 0]$$

$$\left. \begin{aligned} -8 &= 4k \Rightarrow k = -2 \\ 4 &= -2k \Rightarrow k = -2 \\ -2 &= k \end{aligned} \right\} -2 = -2 \Rightarrow \text{totož nebo rovnó}$$

$$A \in p(?) \quad \left. \begin{aligned} -1 &= 3 + 4t \Rightarrow t = -1 \\ 16 &= 14 - 2t \Rightarrow t = -1 \\ 0 &= 1 + t \Rightarrow t = -1 \end{aligned} \right\} -1 = -1 \Rightarrow A \in p \Rightarrow \text{totož}$$

$$2) d: \begin{aligned} x &= 7 - 4s \\ y &= 4s \\ z &= 2s \end{aligned} \quad p: \begin{aligned} x &= 3 + 4t \\ y &= 14 - 2t \\ z &= 1 + t \end{aligned}$$

$$\left. \begin{aligned} -4 &= 4k \\ 4 &= -2k \\ 2 &= k \end{aligned} \right\} -1 \neq 2 \quad \begin{cases} \text{různo} \\ \text{mimo} \end{cases}$$

$$7 - 4s = 3 + 4t$$

$$4s = 14 - 2t$$

$$2s = 1 + t \Rightarrow t = 2s - 1$$

$$7 - 4s = 3 + 4(2s - 1)$$

$$4s = 14 - 2(2s - 1)$$

$$7 - 4s = 3 + 8s - 4$$

$$4s = 14 - 4s + 2$$

$$\left. \begin{aligned} 8 &= 12s \Rightarrow s = \frac{8}{12} = \frac{4}{6} = \frac{2}{3} \\ 16 &= 8s \Rightarrow s = 2 \end{aligned} \right\} \frac{2}{3} \neq 2 \Rightarrow \text{mimo}$$

$$3) \quad p: \begin{cases} x = 2+t \\ y = 1+2t \\ z = 3-t \end{cases} \quad \vec{s}_p = (1, 2, -1) \\ A = [2, 1, 3]$$

$$\rho: 3x - y + z + 1 = 0 \quad \vec{n}_\rho = (3, -1, 1)$$

$$p \parallel \rho (?): \vec{s}_p \cdot \vec{n}_\rho = 0$$

$$(1, 2, -1) \cdot (3, -1, 1) = 0$$

$$3 + (-2) + (-1) = 0$$

$$0 = 0 \Rightarrow p \parallel \rho$$

$$A \in \rho (?): 3 \cdot 2 - 1 + 3 + 1 = 0$$

$$6 - 1 + 3 = 0$$

$$9 \neq 0 \Rightarrow A \notin \rho$$

$$4) \quad M = [1, 0, 5]$$

$$p = \overleftrightarrow{AB}$$

$$A = [0, 1, 0]$$

$$B = [1, 0, 2]$$

$$\left. \begin{matrix} A = [0, 1, 0] \\ B = [1, 0, 2] \end{matrix} \right\} \vec{AB} = (1, -1, 2) = \vec{s}_p = \vec{n}_\rho$$

$$p: x = t$$

$$y = 1 - t$$

$$z = 2t$$

$$\rho: x - y + 2z + d = 0$$

$$M \in \rho \rightarrow 1 - 1 + 10 + d = 0 \\ \underline{d = -11}$$

$$\rho: x - y + 2z - 11 = 0$$

$$t - 1 + t + 4t - 11 = 0$$

$$6t - 12 = 0$$

$$\underline{t = 2}$$

$$P = [2, -1, 4]$$

$$x = 2$$

$$y = 1 - 2$$

$$z = 4$$

$$V = |MP| = \sqrt{1 + 1 + 1} = \underline{\underline{\sqrt{3}}}$$

$$5) \begin{cases} \rho: x - 2y + z + 2 = 0 \\ \sigma: 2x - y - z - 3 = 0 \end{cases} \quad \textcircled{+}$$

$$3x - 3y - 1 = 0$$

$$3x = 3y + 1$$

$$3x = 3t + 1$$

$$x = t + \frac{1}{3}$$

$$\boxed{y = t}$$

$$t + \frac{1}{3} - 2t + z + 2 = 0 \quad / \cdot 3$$

$$3t + 1 - 6t + 3z + 6 = 0$$

$$3z = 3t - 7$$

$$z = t - \frac{7}{3}$$

$$x = t + \frac{1}{3}$$

$$y = t$$

$$z = t - \frac{7}{3}$$